

Chapter 1

Atomic structure

Subatomic particles of the atom:

1. Electron:

Discovered by: J.J. Thomson (1897)

experiment: Cathode Ray Tube

main points:

Cathode rays = electrons, negatively charged particles

They travel in straight lines from cathode $\rightarrow$ anode

They're deflected by electric and magnetic fields $\Rightarrow$ proves they are charged.

can cause fluorescence heat ionization chemical reactions and produce x-rays when striking metals

charge to mass ratio: $e/m = 1.7528 \cdot 10^{11} \text{ C/Kg}$

charge on one electron: $e = -1.602 \cdot 10^{-19} \text{ C}$
(millikan's oil drop experiment 1909)

2. Proton :

Discovered by : Ernest Rutherford (1917)

first observed : E Goldstein (1886) Canal rays / Positive rays

main points :

Positive rays (opposite to cathode rays) = show the presence of positively charged particles

Proton = positively charged particle from hydrogen atom

Charge : $+1.602 \cdot 10^{-19} \text{ C}$

mass : $1.6726 \cdot 10^{-27} \text{ Kg}$ (≈ 1836 times heavier than electron)

3. Neutron :

Discovered by : James Chadwick (1932)

Experiment : Bombarding beryllium with alpha particles

main points :

neutral radiation = neutron (no charge)

Mass $\approx$ same as proton

can penetrate matter deeply

Used in nuclear reactions and cancer treatment

• Rutherford's model of the atom (1911) :

- experiment : Gold foil experiment

shot alpha particles at thin gold foil,

most went straight $\rightarrow$ atom mostly empty space

few deflected $\rightarrow$ atom has a small dense positively charged nucleus

- Model summary :

Nucleus $\rightarrow$ protons (+) + neutrons (neutral)

Electrons move around the nucleus

atom = mostly empty space

(prob : why electrons don't fall into nucleus)

- atomic and mass number :

atomic num $Z \rightarrow$ num of protons = num of electrons (neutral atom)

mass num $A \rightarrow$ num of protons + neutrons $A = Z + N$

Isotopes $\rightarrow$ same element : same Z } $H : H^1, H^2, H^3$
diff A

average atomic mass = $\sum (\text{isotope mass} \times \text{abundance})$
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- Key questions:

- mass of atom mostly comes from $\rightarrow$ protons and neutrons
particles $\approx$ amu mass $\rightarrow$ proton and neutron

- Gold foil experiment shows $\rightarrow$ atom mostly empty, dense nucleus

- proton vs electrons $\rightarrow$ same charge amount, opposite sign

- oxygen atom (8p, 8n, 8e) $\rightarrow$ mass num = 16
 $A = Z + N = 16$

- Rutherford discovered the nucleus Bohr later improves this by explaining electron orbits and energy levels

+ atomic behaviours come from how electrons are arranged.